

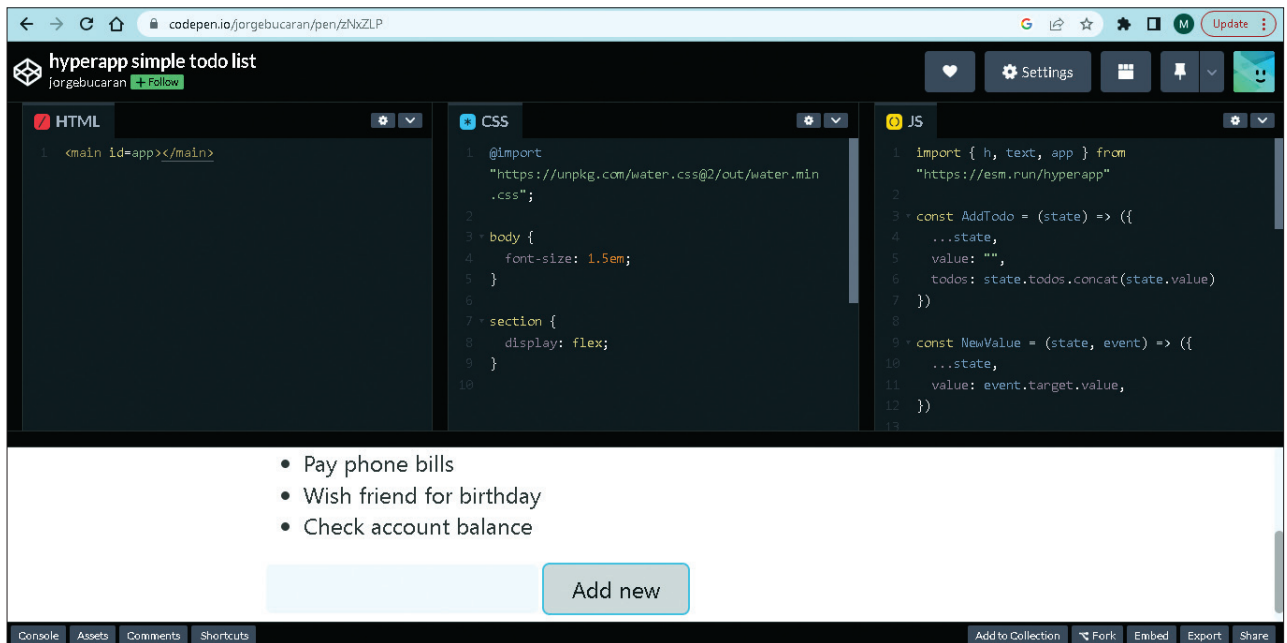


Figure 1: Screenshot of developing a HyperApp application in CodePen editor

3. Set up a basic HyperApp component using JSX syntax.
4. Render the component to the DOM.

HyperApp leverages JSX syntax, allowing developers to write expressive and declarative UI components. It boasts a simple and intuitive API, making it easy for developers to grasp and leverage its capabilities. The library employs a declarative approach, where developers describe the desired user interface state and let HyperApp handle the rendering and updates. This approach promotes code clarity and maintainability, allowing developers to focus on building application logic rather than worrying about the intricacies of the underlying framework.

HyperApp leverages a virtual DOM (VDOM) implementation to efficiently update and render UI components. By comparing the VDOM with the actual DOM, HyperApp minimises unnecessary re-renders, resulting in optimal performance. The library's efficient Diffing algorithm ensures that only the necessary updates are applied, reducing rendering overhead and enabling fast and responsive user interfaces.

Reactive state management

HyperApp incorporates a reactive state management system that simplifies data handling within applications. It provides a central store, inspired by Redux, where the application state is managed. Any changes to the state trigger automatic re-rendering of relevant components, ensuring a consistent and up-to-date UI. HyperApp's reactive approach allows for easier state management and promotes the development of scalable and maintainable applications.

Architecture of HyperApp

HyperApp embraces a modular architecture, allowing developers to selectively include features and extensions based on their project requirements. The library provides a rich ecosystem of plugins and community-contributed packages, offering additional functionalities such as routing, HTTP requests, and form handling. This modular approach promotes code reusability and empowers developers to customise the development environment based on their specific needs.

With its lightweight design and efficient rendering mechanism, HyperApp delivers exceptional performance, even in complex applications. The library's focus on minimalism and performance optimisation enables faster page loads, smoother transitions, and improved user experiences. HyperApp's scalability is evident in its ability to handle large data sets and complex application logic while maintaining optimal performance.

HyperApp benefits from an active and growing community that contributes to its ecosystem with plugins, tools, and resources. Its community-driven nature ensures ongoing support, updates, and documentation, making it easier for developers to get started and collaborate.

With HyperApp, developers can create responsive and scalable applications while maintaining code clarity and simplicity. As the ecosystem continues to grow and evolve, HyperApp is poised to play an increasingly significant role in the development of performant web applications.